

E-COMMERCE INTELLIGENCE & DECISION SUPPORT PLATFORM

Initial Project Overview

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Project Title

E-Commerce Intelligence & Decision Support Platform

Problem Statement

E-commerce managers generate large volumes of transaction data but struggle to turn it into clear decisions about customers and products. This project builds one platform where a manager can explore customer and product behaviour and get personalized product recommendations, all from historical data.

Proposed System

A full-stack web application for an e-commerce manager, with one login, analytics dashboards, and a recommendation service, all backed by a shared database and ML model. The core ML task predicts customer-product purchase likelihood and ranks products into Top-5 recommendations, evaluated with Hit-rate@5 against a random and a popularity baseline. Statistics support the feature engineering and evaluation; Power BI/Tableau turns the results into a management-facing report.

Objectives

- Build one full-stack platform (frontend, backend, database, ML) rather than separate subject-wise projects.
- Clean the Online Retail II dataset and engineer customer/product behaviour features.
- Train a purchase-likelihood model and generate Top-5 recommendations.
- Evaluate with Hit-rate@5 against Level 0 (random) and Level 1 (popularity) baselines.
- Apply statistics (distributions, correlation, chi-square, confidence intervals, cross-validation) and standard SE practice (SRS, APIs, testing).

Scope

A focused prototype covering the full path from raw transaction data to a usable recommendation, kept lean enough to finish solo in 90 hours (~20 for modelling). Roughly 8 screens: login, executive dashboard (with cancellations), customer intelligence (with segments), product intelligence (with country view), recommendation workspace, and model performance — plus the Power BI/Tableau report.

Main User Workflow

- Log in and view the executive dashboard.

- Open a customer profile to see behaviour and segment.
- Generate Top-5 recommendations for that customer.
- Check product/country analytics and model performance.
- Use Power BI/Tableau for management-level drill-down.

Main Areas Covered

- **Machine Learning:** Logistic Regression (primary) vs. Decision Tree; K-Means for segmentation.
- **Statistics:** distributions, correlation, chi-square, confidence intervals, cross-validation.
- **Data Visualization:** Plotly dashboards plus Power BI/Tableau reporting.
- **Software Engineering:** SRS, database design, APIs, testing, documentation.
- **Full-Stack:** React, FastAPI, MySQL/PostgreSQL, REST APIs, ML inference service.

Expected Outcomes

- A working full-stack recommendation platform with analytics and reporting.
- A leakage-safe purchase-likelihood model evaluated against required baselines.
- Complete SE documentation: SRS, design, testing evidence.

Planned Dataset

UCI Online Retail II — archive.ics.uci.edu/dataset/502/online+retail+ii — ~1,067,371 transaction-line records (Dec 2009–Dec 2011). Each row is one invoice line, not one order, so features must be built by aggregation.

Project Direction

Data → Clean → Statistics → Features → ML Model → Top-5 Recommendation → API → Frontend → Dashboards → Testing.